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Example 1.1. The differential equation $y' = y^2$ is a separable equation. It can be solved by the method of separation of variables. The general solution is $y = \frac{1}{C - x}$, where C is an arbitrary constant.

Example 1.2. The differential equation $y' = y^2 + 1$ is a separable equation. It can be solved by the method of separation of variables. The general solution is $y = \tan(x + C)$, where C is an arbitrary constant.

1.1 The general solution of a separable equation

A differential equation is called separable if it can be written in the form $y' = f(x)g(y)$, where $f(x)$ and $g(y)$ are functions of x and y respectively. In this case, the equation can be solved by the method of separation of variables. The general solution is $y = \int \frac{f(x)}{g(y)} dx + C$, where C is an arbitrary constant.

1.2 The method of separation of variables

The method of separation of variables is a technique for solving differential equations. It involves separating the variables x and y in the equation, so that the equation can be written in the form $y' = f(x)g(y)$. The general solution is then found by integrating both sides of the equation.

$$\begin{array}{c} \text{separable} \\ \hline \begin{array}{l} y' = f(x)g(y) \\ y' = f(x) \\ y' = g(y) \\ y' = f(x) + g(y) \\ y' = f(x)g(y) \end{array} \end{array}$$

Example 1.3. The differential equation $y' = y^2 + 1$ is a separable equation. It can be solved by the method of separation of variables. The general solution is $y = \tan(x + C)$, where C is an arbitrary constant.

$$y' = f(x)g(y) \quad \text{for } f(x) = \frac{1}{x^2} \quad g(y) = y^2 + 1$$

The differential equation $y' = y^2 + 1$ is a separable equation. It can be solved by the method of separation of variables. The general solution is $y = \tan(x + C)$, where C is an arbitrary constant.

Example 1.4. The differential equation $y' = y^2 + 1$ is a separable equation. It can be solved by the method of separation of variables. The general solution is $y = \tan(x + C)$, where C is an arbitrary constant.